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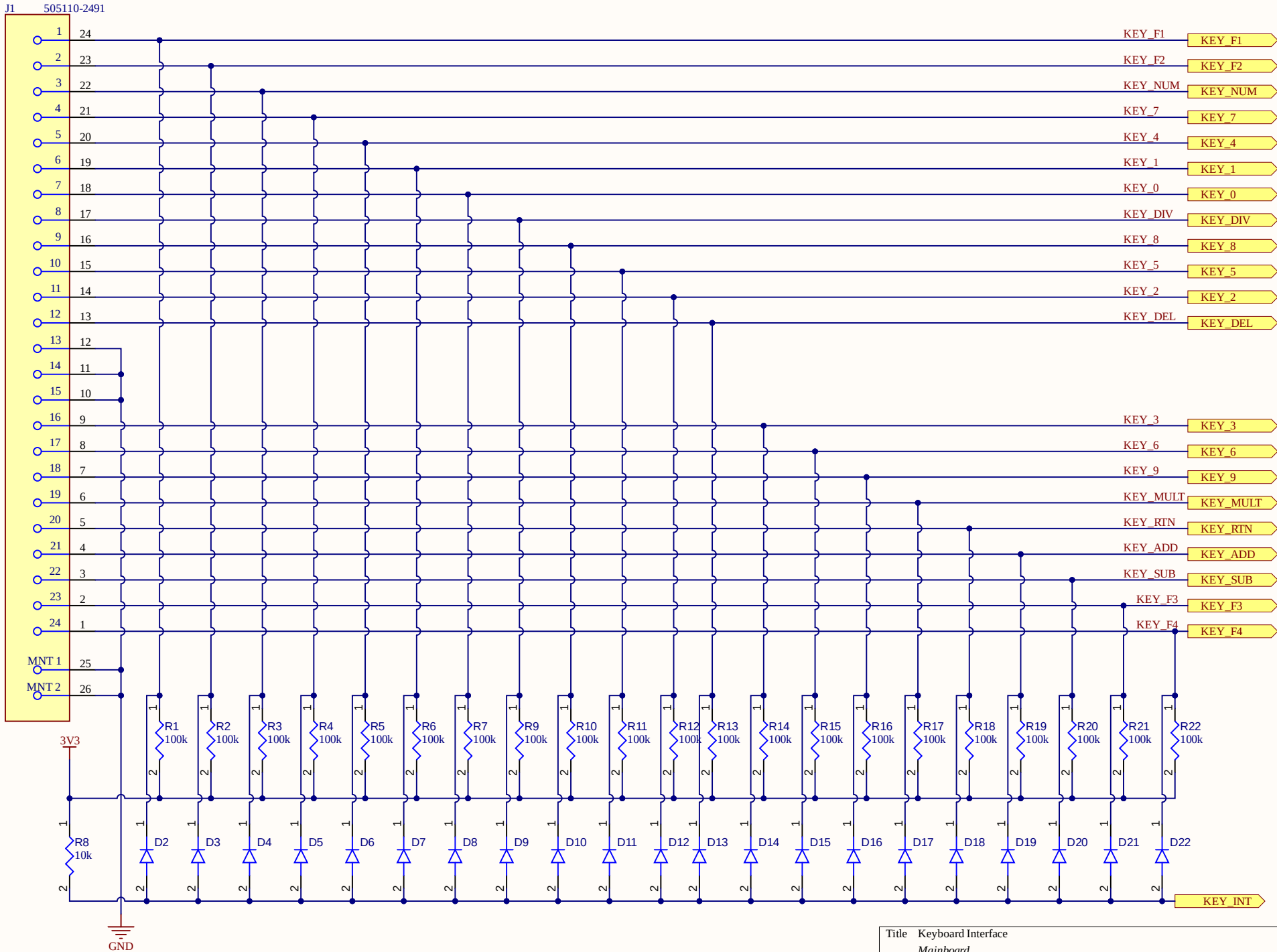
D

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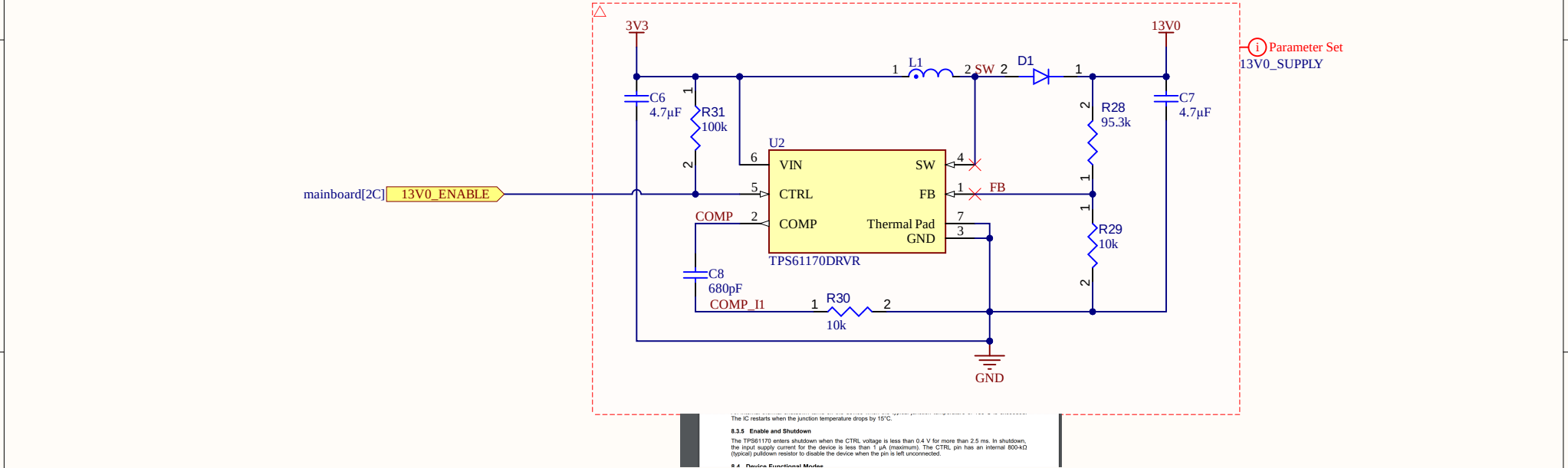
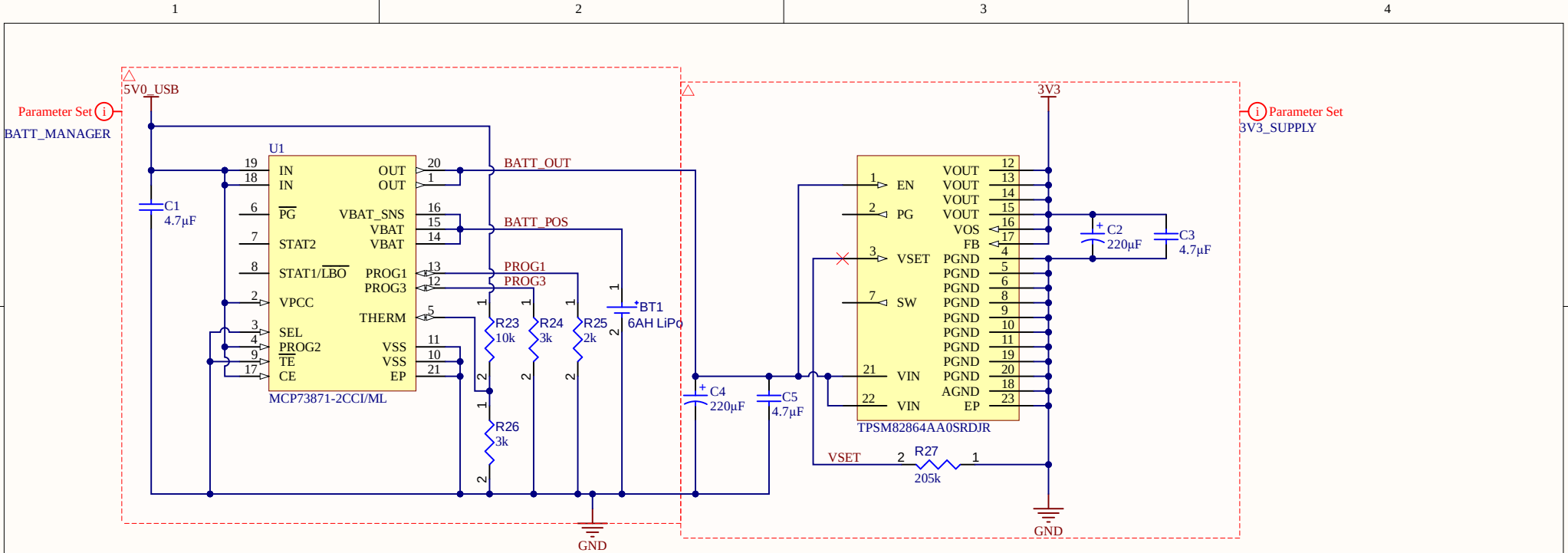
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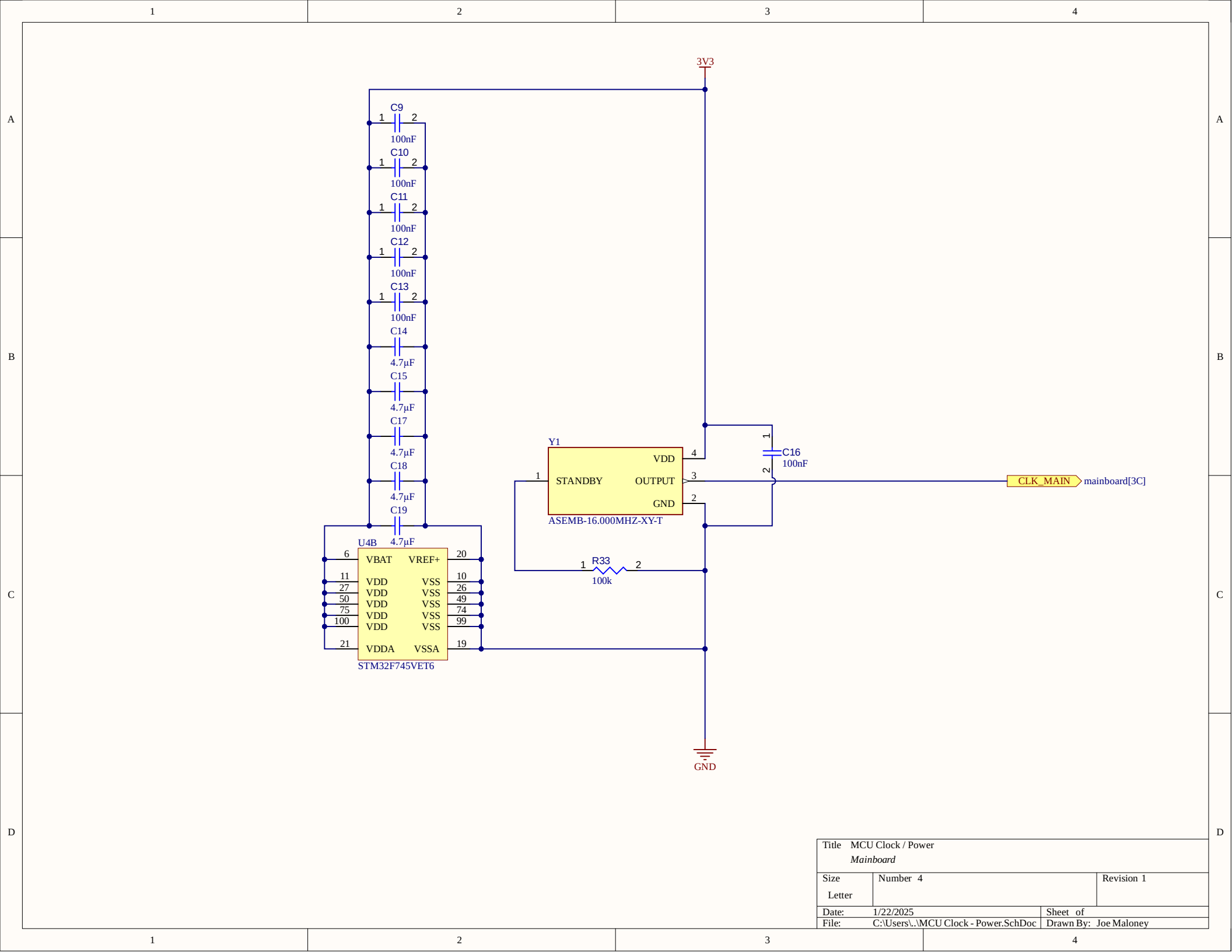
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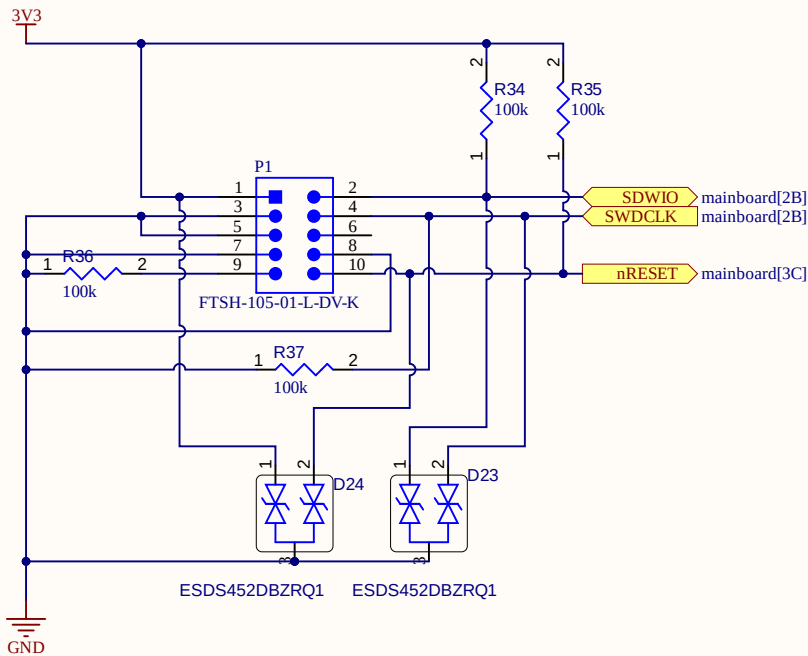


Title Keyboard Interface Mainboard		
Size Letter	Number 2	Revision 1
Date:	1/22/2025	Sheet of
File:	C:\Users\Keyboard Interface.SchDoc	Drawn By: Joe Maloney



Title Power Supplies Mainboard		
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Serial Wire Signals

The Serial Wire mode differs to JTAG debugging, because only two pins are used for the communication. A third pin can be used optionally to trace data. JTAG pins and SW pins are shared.

- **TCLK** is **SWCLK** (Serial Wire Clock)
- **TMS** is **SWDIO** (Serial Wire debug Data Input/Output)
- **TDO** is **SWO** (Serial Wire trace Output)

Signal	Connects to...
SWDIO	Data I/O pin -- Use 100k Ohm pull-up resistor to VCC.
SWO	Optional trace output pin.
SWCLK	Clock pin -- Use 100k Ohm pull-down resistor to GND.
VCC	Positive Supply Voltage -- Power supply for JTAG interface drivers.
GND	Digital ground.
RESET	RSTIN/ pin -- Connect this pin to the (active low) reset input of the target CPU.

Note:

- Usually, devices do not include pull-up or pull-down resistors on JTAG nor SW pins. Resistors should be added externally onto the board. However, do not add resistors when the device includes them already.

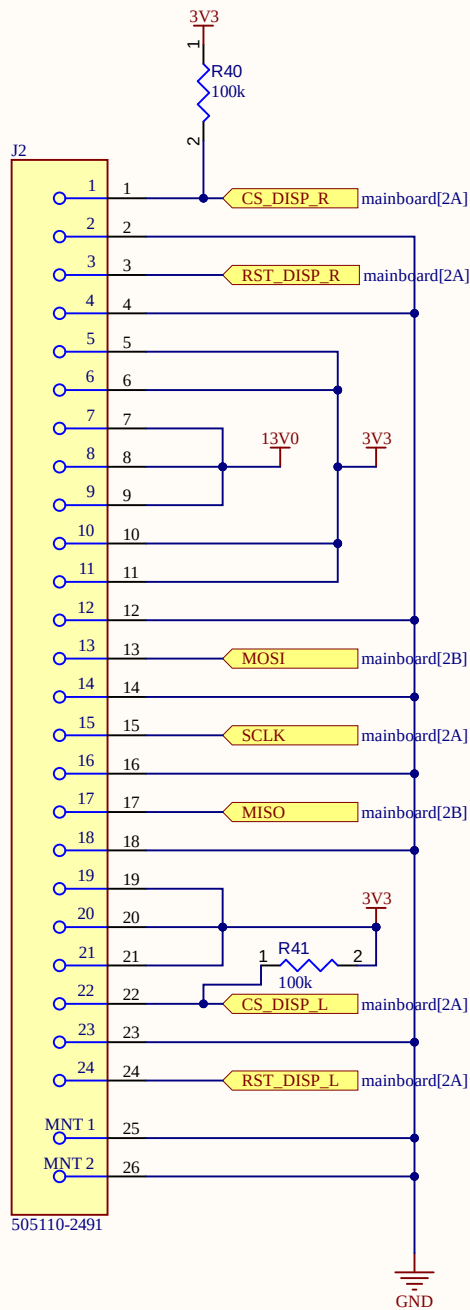
JTAG/SWD Connector

The 10-pin, 0.05" JTAG/SWD connector offers ITM and DWT trace information. In SW and the second pin (SWDCLK) clocks the data. A third pin (SWO) delivers the trace data.

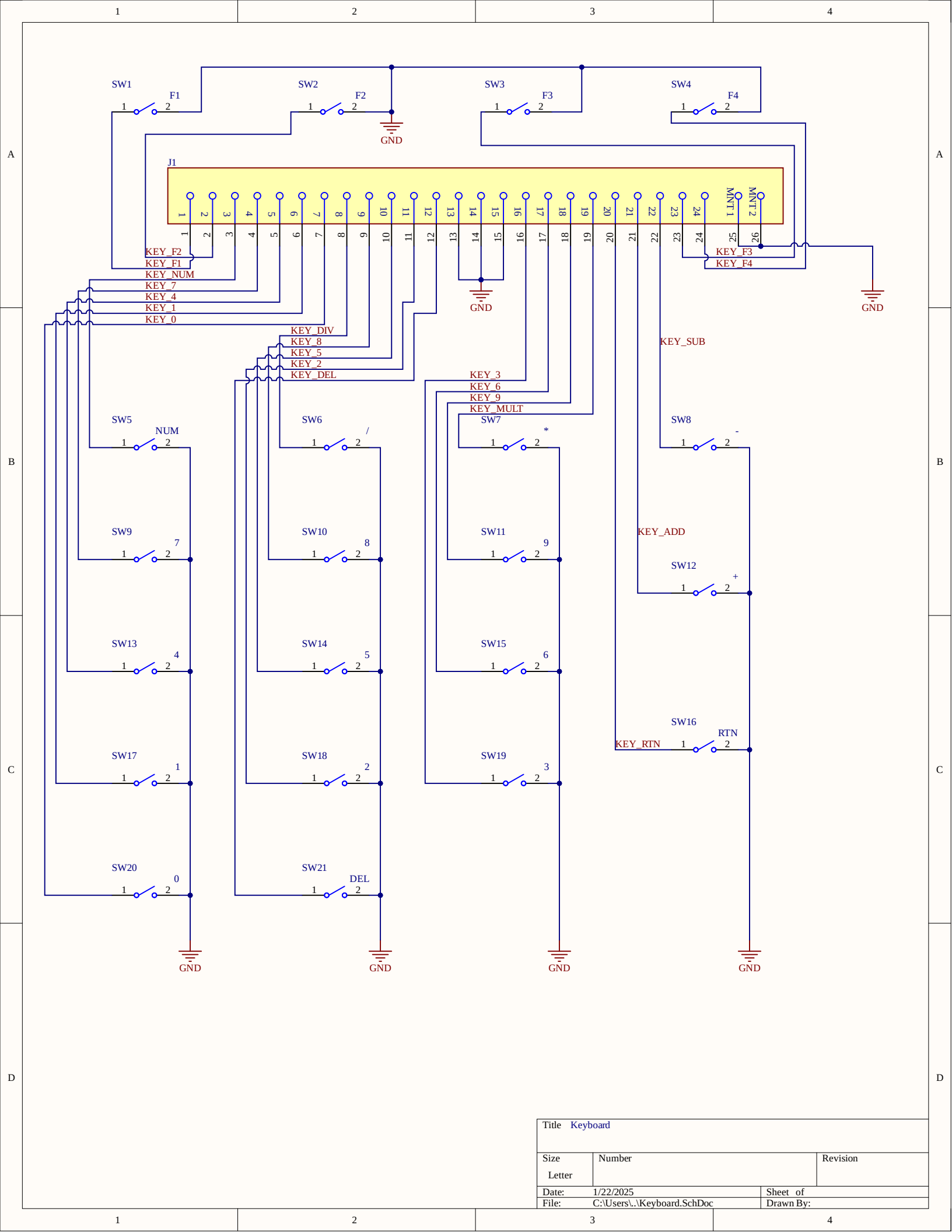
10-pin JTAG/SWD Connector

VCC	1	☐	2	☐	SDWIO / TMS
GND	3	☐	4	☐	SWDCLK / TCLK
GND	5	☐	6	☐	SWO / TDO
KEY	7	☐	8	☐	NC / TDI
GNDDetect	9	☐	10	☐	nRESET

Title SWD Mainboard		
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Title Display Mainboard		
Size Letter	Number 6	Revision 1
Date:	1/22/2025	Sheet of
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Title Keyboard		
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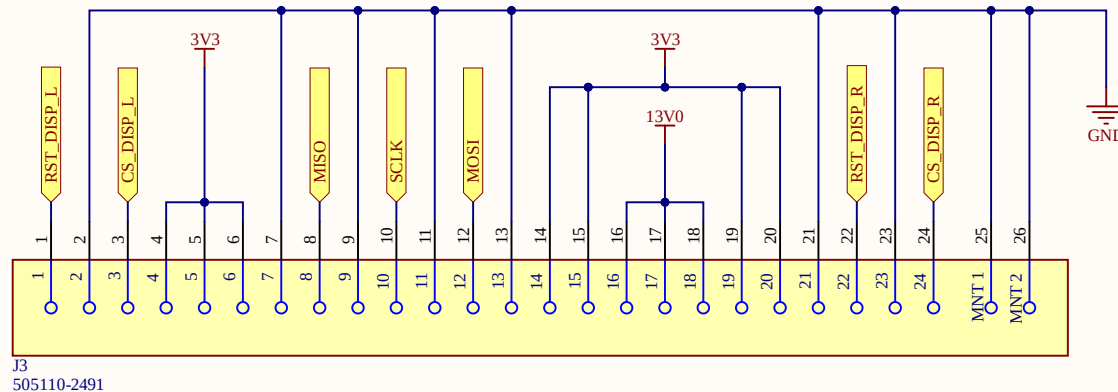
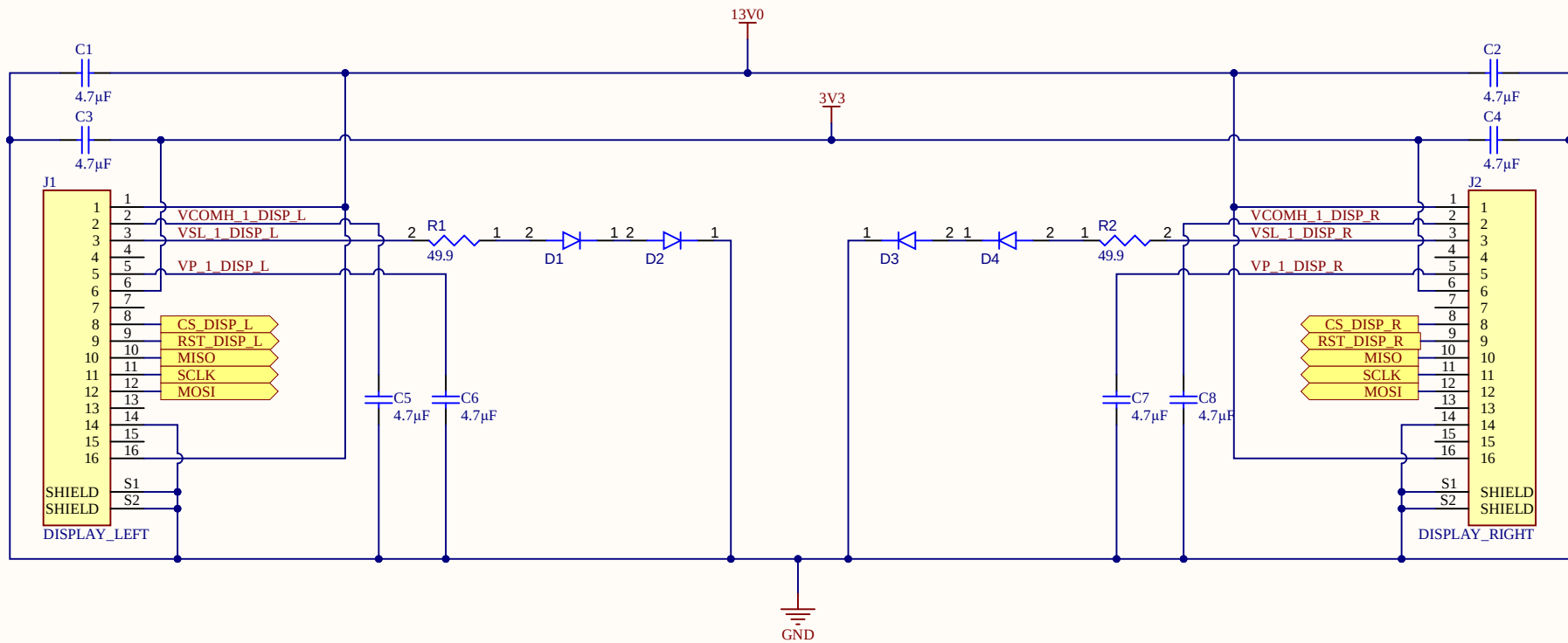
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Title Displayboard Displayboard		
Size Letter	Number 1	Revision 1.0
Date:	1/22/2025	Sheet of
File:	C:\Users\...\Displayboard.SchDoc	Drawn By: Joe Maloney